System Formulation: Cosserat Rod Dynamics

Well-Posedness via the ($\mathcal{L}$, Ω, BC, IC, $\boldsymbol{\theta}$) Tuple

Snake-HRL Project — Reference Document

Nomenclature

Coordinates and domain.

s	Arc-length coordinate along the rod, $s \in [0, L]$
t	Time, $t \in [0, T]$
L	Total rod length (1.0 m)
T	Simulation horizon (episode-dependent)
Ω	Spatio-temporal domain: $\{(s,t) \mid s \in [0,L],\ t \in [0,T]\}$
$\partial\Omega$	Boundary of the domain

State variables (continuous).

$\mathbf{x}(s,t)$	Rod centerline position $\in \mathbb{R}^3$
$\mathbf{v}(s,t)$	Translational velocity, $\mathbf{v} = \partial\mathbf{x}/\partial t$
$\mathbf{d}(s,t)$	Material director frame (local orientation triad)
$\boldsymbol{\omega}(s,t)$	Angular velocity of the material frame
$\boldsymbol{\varepsilon}(s,t)$	Generalized strain (shear and axial stretch)
$\boldsymbol{\kappa}(s,t)$	Generalized curvature (bending and twist)
$\boldsymbol{\kappa}_0(s,t)$	Rest curvature — the control input from the CPG
$\boldsymbol{\sigma}(s,t)$	Generalized strain in the local (material) frame

Discrete state variables (semi-discrete / ODE).

x_i, y_i	Nodal positions in the plane, $i = 1, \ldots, N$
$\dot{x}_i, \dot{y}_i$	Nodal translational velocities
$\ddot{x}_i, \ddot{y}_i$	Nodal translational accelerations (not stored as state)
ψ_e	Element yaw angle, $e = 1, \ldots, N_e$
$\omega_{z,e}$	Element angular velocity about the vertical (z) axis
$\dot{\omega}_{z,e}$	Element angular acceleration (not stored as state)
$\mathbf{s}$	Full state vector $\in \mathbb{R}^{124}$: $(x_i, y_i, \dot{x}_i, \dot{y}_i, \psi_e, \omega_{z,e})$
$\mathbf{q}$	Generalized positions: $(x_i, y_i, \psi_e) \in \mathbb{R}^{62}$
$\dot{\mathbf{q}}$	Generalized velocities: $(\dot{x}_i, \dot{y}_i, \omega_{z,e}) \in \mathbb{R}^{62}$

Internal forces and moments.

$\mathbf{F}(s,t)$	Internal force resultant (shear + tension)
$\mathbf{M}(s,t)$	Internal moment resultant (bending + torsion)
$\mathbf{n}_e^L$	Internal stress on element e in the local frame
$\boldsymbol{\tau}_v^L$	Bending–twist couple at Voronoi node v
$\mathbf{F}_i^{\text{int}}$	Net internal force on node i (from finite differences)
$\mathbf{m}_e^{\text{int}}$	Net internal torque on element e (five-term assembly)

External forces.

$\mathbf{f}_{\text{ext}}$	External distributed force (continuous)
$\mathbf{m}_{\text{ext}}$	External distributed moment (continuous)
$\mathbf{f}_i^{\text{grav}}$	Gravitational force on node i: $m_i\mathbf{g}$
$\mathbf{f}_i^{\text{rft}}$	Anisotropic ground friction (RFT) on node i
$\mathbf{f}_i^{\text{damp}}$	Numerical damping force on node i: $-\gamma\,\mathbf{v}_i$
$\mathbf{F}^{\text{ext}}$	Total external force vector (all nodes, stacked)
$\mathbf{m}_e^{\text{ext}}$	External torque on element e

Material properties and constitutive law.

E	Young's modulus (2×10^6 Pa)
G	Shear modulus, $G = E/2(1+\nu)$
ν	Poisson's ratio (0.5)
ρ	Material density ($1200\,\mathrm{kg\,m^{-3}}$)
R	Cross-sectional radius (0.001 m)
A	Cross-sectional area, $A = \pi R^2$
I	Second moment of area, $I = \pi R^4/4$
J	Polar second moment of area, $J = \pi R^4/2$
$\mathbf{S}$	Shear–stretch stiffness matrix: $\mathrm{diag}(GA, GA, EA)$
$\mathbf{B}$	Bending–twist stiffness matrix: $\mathrm{diag}(EI, EI, GJ)$
EA, GA	Axial and shear stiffness (scalar products)
EI, GJ	Bending and torsional stiffness (scalar products)

Geometry and discretization.

N	Number of nodes (21)
N_e	Number of elements (20), $N_e = N - 1$
m_i	Lumped mass at node i
$\mathbf{J}_e$	Second moment of inertia tensor of element e
ℓ_e	Current length of element e
ℓ_e^{rest}	Rest (reference) length of element e
ε_e	Dilatation (axial stretch ratio): $\ell_e/\ell_e^{\text{rest}}$

$\hat{\varepsilon}_v$	Voronoi dilatation at Voronoi node v
$\dot{\varepsilon}_e$	Dilatation rate (time derivative of ε_e)
$\mathbf{t}_e$	Unit tangent vector of element e
κ_e	Discrete curvature at element e
κ_e^{rest}	Rest curvature at element e (set by CPG)
$\boldsymbol{\kappa}_v$	Curvature vector at Voronoi node v

Friction and damping.

c_t	Tangential friction coefficient (0.01)
c_n	Normal friction coefficient (0.1)
γ	Numerical damping coefficient (0.1)
$\mathbf{v}_{t,i}$	Tangential component of velocity at node i
$\mathbf{v}_{n,i}$	Normal component of velocity at node i
$\mathbf{g}$	Gravitational acceleration vector, $(0, 0, -9.81)\,\mathrm{m\,s^{-2}}$

Time integration.

Δt	Integration substep size (0.001 s)
Δt_{ctrl}	RL control step duration (0.5 s)
n_{sub}	Number of substeps per control step (500)
$V_{\Delta t}$	One Position Verlet substep map: $\mathbb{R}^{124} \to \mathbb{R}^{124}$
$T(\mathbf{s}_t, \mathbf{a}_t)$	Transition operator: composition of n_{sub} substeps
n	Substep index (superscript)

Matrix operators (Section 7).

$\mathbf{D}$	Difference operator (node $\to$ element), $\mathbb{R}^{20\times 21}$, bidiagonal
$\mathbf{D}^T$	Transpose difference (element $\to$ node), force assembly
$\mathbf{A}_v$	Voronoi averaging (element $\to$ Voronoi), $\mathbb{R}^{19\times 20}$, tridiagonal
$\mathbf{D}_v$	Voronoi difference (Voronoi $\to$ element), $\mathbb{R}^{20\times 19}$, bidiagonal
$\mathbf{S}_{\text{blk}}$	Block-diagonal shear–stretch stiffness, $\mathbb{R}^{60\times 60}$
$\mathbf{B}_{\text{blk}}$	Block-diagonal bending–twist stiffness, $\mathbb{R}^{57\times 57}$
$\mathbf{M}^{-1}$	Diagonal inverse mass matrix, $\mathbb{R}^{21\times 21}$
$\mathbf{J}_{\text{blk}}^{-1}$	Block-diagonal inverse inertia, $\mathbb{R}^{60\times 60}$
$\mathbf{Q}_{\text{blk}}(\boldsymbol{\psi})$	Block-diagonal rotation (state-dependent), $\mathbb{R}^{40\times 40}$
$\mathbf{Q}_e$	2×2 rotation matrix for element e: entries $\cos\psi_e, \sin\psi_e$
$\mathbf{C}_{\text{rft}}(\mathbf{t})$	Friction projection matrix (state-dependent), $\mathbb{R}^{42\times 42}$
$\mathbf{P}_t$	Tangential projection: $\mathrm{blkdiag}(\mathbf{t}_i\,\mathbf{t}_i^T)$
$\mathbf{I}_k$	Identity matrix of size k

Element-wise operations.

$\|\cdot\|$	Euclidean norm (per element or node)
$\odot$	Hadamard (element-wise) product
$\oslash$	Element-wise division
$\times$	Cross product (bilinear, per element)
Δ_s	Discrete spatial derivative (difference kernel on staggered grid)
$\mathcal{A}[\cdot]$	Quadrature (averaging) kernel on Voronoi grid
$\bigoplus$	Direct sum (block-diagonal assembly)

Control (CPG action).

$\mathbf{a}$	RL action vector $\in \mathbb{R}^5$: (A, f, k, ϕ_0, b)
A	CPG amplitude (rad m^{-1})
f	CPG frequency (Hz)
k	CPG wave number (wavelengths per body length)
ϕ_0	CPG phase offset (rad)
b	CPG turn bias (rad m^{-1})

System-level notation.

$\mathcal{S}$	The well-posed system: $(\mathcal{L}, \Omega, \mathrm{BC}, \mathrm{IC}, \boldsymbol{\theta})$
$\mathcal{S}_h$	Semi-discrete system (after spatial discretization)
$\mathcal{L}$	Differential operator (governing equations)
$\mathbf{g}(\mathbf{s}, \mathbf{a}, t)$	Right-hand side of the semi-discrete ODE system
$\boldsymbol{\theta}$	Parameter vector (material, friction, damping, solver)
f_θ	Neural surrogate model: $\mathbb{R}^{129} \to \mathbb{R}^{124}$
$\mathbf{z}_t$	Surrogate input (state + action concatenation)

Formatting conventions.

$(\cdot)^{n+1/2}$	Value at half-step
$(\cdot)^{\mathrm{rest}}$	Reference / undeformed value
$(\cdot)^{\mathrm{int}}$	Internal (elastic) contribution
$(\cdot)^{\mathrm{ext}}$	External contribution
$(\cdot)^{L}$	Quantity expressed in the local (material) frame

1 The Well-Posedness Tuple

A well-posed initial–boundary value problem (IBVP) is uniquely determined by five components (Hadamard's conditions):

$$\boxed{\mathcal{S} = \big(\mathcal{L},\ \ \Omega,\ \ \text{BC},\ \ \text{IC},\ \ \boldsymbol{\theta}\big)} \tag{1}$$

Symbol	Name	Meaning
$\mathcal{L}$	Differential operator	The governing equation(s) — the relationship between unknowns and their derivatives
Ω	Domain	Where the equations hold (spatial region $\times$ time interval)
BC	Boundary conditions	Constraints at the boundary $\partial\Omega$
IC	Initial conditions	The system state at $t = 0$
$\boldsymbol{\theta}$	Parameters	Physical constants, material properties, and forcing terms

Changing *any one* of these five components yields a different system with a different solution. A problem is well-posed when these components are specified such that a solution (i) exists, (ii) is unique, and (iii) depends continuously on the data.

2 Continuous Cosserat Rod System

The snake robot body is modeled as a Cosserat rod — a one-dimensional elastic continuum that supports shear, extension, bending, and torsion. The governing equations are the conservation of linear and angular momentum per unit arc length.

2.1 $\mathcal{L}$ — Governing PDEs

$$\rho A\,\frac{\partial \mathbf{v}}{\partial t} = \frac{\partial \mathbf{F}}{\partial s} + \mathbf{f}_{\text{ext}} \qquad \text{(linear momentum)} \tag{2a}$$

$$\rho \mathbf{I}\,\frac{\partial \boldsymbol{\omega}}{\partial t} = \frac{\partial \mathbf{M}}{\partial s} + \mathbf{m}_{\text{ext}} + \frac{\partial \mathbf{x}}{\partial s} \times \mathbf{F} \qquad \text{(angular momentum)} \tag{2b}$$

with kinematic relations $\mathbf{v} = \partial \mathbf{x}/\partial t$ (translational velocity) and $\boldsymbol{\omega}$ (angular velocity of the material frame), and constitutive (closure) relations:

$$\mathbf{F} = \mathbf{S}\,(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0), \qquad \mathbf{S} = \operatorname{diag}(GA,\, GA,\, EA) \tag{3a}$$

$$\mathbf{M} = \mathbf{B}\,(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0), \qquad \mathbf{B} = \operatorname{diag}(EI,\, EI,\, GJ) \tag{3b}$$

- $\mathbf{x}(s,t) \in \mathbb{R}^3$ — rod centerline position at arc length s
- $\mathbf{d}(s,t)$ — material director frame (local orientation)

- $\mathbf{F}(s,t)$ — internal force resultant (shear + tension)
- $\mathbf{M}(s,t)$ — internal moment resultant (bending + torsion)
- $\boldsymbol{\varepsilon}$ — generalized strain (shear and axial stretch)
- $\boldsymbol{\kappa}$ — generalized curvature (bending and twist)
- $\boldsymbol{\kappa}_0(s,t)$ — **rest curvature** (the control input from the CPG)
- $\mathbf{f}_{\text{ext}}$ — external distributed force (gravity, RFT ground friction)
- $\mathbf{m}_{\text{ext}}$ — external distributed moment (CPG muscle torque)

2.2 Ω — Domain

$$\Omega = \left\{ (s,t) \;\middle|\; s \in [0,L], \;\; t \in [0,T] \right\} \tag{4}$$

Symbol	Description		Value
s	Arc-length coordinate	spatial	$[0, 1.0]$ m
L	Total rod length		1.0 m
t	Time coordinate	temporal	$[0, T]$
T	Simulation horizon		episode-dependent

2.3 BC — Boundary Conditions

The snake rod is **free–free** (no clamped or pinned ends):

$$\mathbf{F}(0,t) = \mathbf{0}, \qquad \mathbf{F}(L,t) = \mathbf{0} \qquad \text{(no end forces)} \tag{5a}$$

$$\mathbf{M}(0,t) = \mathbf{0}, \qquad \mathbf{M}(L,t) = \mathbf{0} \qquad \text{(no end moments)} \tag{5b}$$

Both endpoints are traction-free: no external force or moment is applied at either end of the rod. The snake is free to translate and rotate in the plane.

2.4 IC — Initial Conditions

$$\mathbf{x}(s,0) = \mathbf{x}_0 + s\,\hat{\mathbf{e}}_x \qquad \text{(straight rod along } x\text{-axis)} \tag{6a}$$

$$\mathbf{v}(s,0) = \mathbf{0} \qquad \text{(at rest)} \tag{6b}$$

$$\boldsymbol{\kappa}(s,0) = \mathbf{0} \qquad \text{(zero curvature — straight)} \tag{6c}$$

$$\boldsymbol{\omega}(s,0) = \mathbf{0} \qquad \text{(no rotation)} \tag{6d}$$

where $\mathbf{x}_0$ is the starting position of the snake head.

2.5 θ — Parameters

The parameter vector is organized into four groups.

Symbol	Name	Config field	Value	Units
E	Young's modulus	`youngs_modulus`	2×10^6	Pa
ν	Poisson's ratio	`poisson_ratio`	0.5	—
G	Shear modulus	$E/2(1+\nu)$	6.67×10^5	Pa
ρ	Density	`density`	1200	$\mathrm{kg\,m^{-3}}$
R	Cross-sectional radius	`snake_radius`	0.001	m

Material properties.

Derived stiffnesses. From the material properties and circular cross-section ($A = \pi R^2$, $I = \pi R^4/4$, $J = \pi R^4/2$):

Symbol	Expression	Physical meaning
EA	$E \cdot \pi R^2$	Axial (stretch) stiffness
GA	$G \cdot \pi R^2$	Shear stiffness
EI	$E \cdot \pi R^4/4$	Bending stiffness
GJ	$G \cdot \pi R^4/2$	Torsional stiffness

Force	Model	Parameters
Gravity	$\mathbf{f}_g = \rho A\,\mathbf{g}$	$\mathbf{g} = (0, 0, -9.81)\,\mathrm{m\,s^{-2}}$
Ground friction (RFT)	$\mathbf{f}_{\text{rft}} = -c_t \mathbf{v}_t - c_n \mathbf{v}_n$	$c_t = 0.01$, $c_n = 0.1$
Numerical damping	$\mathbf{f}_{\text{damp}} = -\gamma\,\mathbf{v}$	$\gamma = 0.1$

External forces.

Symbol	Name	Config field	Value	Units
Δt_{ctrl}	RL control step	`substeps_per_action` × `dt_substep`	0.5	s
Δt	Integration substep	`dt_substep`	0.001	s
n_{sub}	Substeps per control step	`substeps_per_action`	500	—
—	Integrator	`elastica_time_stepper`	Position Verlet	

Time integration.

3 Semi-Discrete ODE System

Spatial discretization with $N = 21$ nodes and $N_e = 20$ elements on a staggered grid replaces the continuous PDEs (2) with a system of 124 coupled first-order ODEs. Restricting to 2D planar motion ($z \equiv 0$):

Nodal translational dynamics $(i = 1, \ldots, N)$:

$$\frac{dx_i}{dt} = \dot{x}_i, \qquad \frac{dy_i}{dt} = \dot{y}_i \tag{7a}$$

$$m_i \frac{d\dot{x}_i}{dt} = \Delta_s F_{x,i} + f_{x,i}^{\text{ext}}, \qquad m_i \frac{d\dot{y}_i}{dt} = \Delta_s F_{y,i} + f_{y,i}^{\text{ext}} \tag{7b}$$

Element rotational dynamics $(e = 1, \ldots, N_e)$:

$$\frac{d\psi_e}{dt} = \omega_{z,e} \tag{8a}$$

$$J_e \frac{d\omega_{z,e}}{dt} = \Delta_s M_{z,e} + m_{z,e}^{\text{ext}} + \left(\Delta_s \mathbf{x}_e \times \mathbf{F}_e\right)_z \tag{8b}$$

where ψ_e is the element yaw angle and $\omega_{z,e}$ is the angular velocity about the vertical axis. The bending moment is given by $M_{z,e} = EI\,(\kappa_e - \kappa_e^{\text{rest}})$, with κ_e^{rest} set by the CPG controller.

State vector. These 124 scalar ODEs define the state vector $\mathbf{s} \in \mathbb{R}^{124}$:

$$\mathbf{s} = \big(\underbrace{x_1, \ldots, x_{21}}_{21},\ \underbrace{y_1, \ldots, y_{21}}_{21},\ \underbrace{\dot{x}_1, \ldots, \dot{x}_{21}}_{21},\ \underbrace{\dot{y}_1, \ldots, \dot{y}_{21}}_{21},\ \underbrace{\psi_1, \ldots, \psi_{20}}_{20},\ \underbrace{\omega_{z,1}, \ldots, \omega_{z,20}}_{20}\big) \tag{9}$$

In compact form, the semi-discrete ODE system is:

$$\boxed{\dot{\mathbf{s}} = \mathbf{g}(\mathbf{s},\, \mathbf{a},\, t), \qquad \mathbf{s} \in \mathbb{R}^{124},\ \ \mathbf{a} \in \mathbb{R}^5} \tag{10}$$

4 The Semi-Discrete System as a Tuple

After spatial discretization, the IBVP becomes an initial value problem (IVP) for an ODE system. The tuple specializes as follows:

$$\boxed{\mathcal{S}_h = \big(\,\mathbf{g},\ \ [0, T],\ \ \text{—},\ \ \mathbf{s}_0,\ \ \boldsymbol{\theta}\,\big)} \tag{11}$$

Component	Specialization	Description
$\mathcal{L} \to \mathbf{g}$	RHS function	124 coupled ODEs (7–8) assembling elastic, contact, and control forces
$\Omega \to [0, T]$	Time interval	Spatial domain absorbed into DOF indexing
BC $\to$ —	Eliminated	Free–free BCs become natural conditions in the discrete force assembly
IC $\to \mathbf{s}_0$	Initial state	$\mathbf{s}_0 \in \mathbb{R}^{124}$: straight rod at rest
$\boldsymbol{\theta}$	Unchanged	Material, friction, damping, and solver parameters

5 Time Integration: The Transition Operator

The Position Verlet scheme advances the ODE system (10) through n_{sub} substeps per control step. Each substep $V_{\Delta t}$ is a second-order symplectic map:

$$\mathbf{q}^{n+\frac{1}{2}} = \mathbf{q}^n + \tfrac{1}{2}\,\Delta t\;\dot{\mathbf{q}}^n \tag{12a}$$

$$\dot{\mathbf{q}}^{n+1} = \dot{\mathbf{q}}^n + \Delta t\;\ddot{\mathbf{q}}(\mathbf{q}^{n+\frac{1}{2}}) \tag{12b}$$

$$\mathbf{q}^{n+1} = \mathbf{q}^{n+\frac{1}{2}} + \tfrac{1}{2}\,\Delta t\;\dot{\mathbf{q}}^{n+1} \tag{12c}$$

Here $\mathbf{q}$ collects the generalized positions (x_i, y_i, ψ_e) and $\dot{\mathbf{q}}$ collects the velocities $(\dot{x}_i, \dot{y}_i, \omega_{z,e})$.

The ground-truth transition operator for one RL control step is the composition:

$$T(\mathbf{s}_t, \mathbf{a}_t) = \underbrace{V_{\Delta t} \circ V_{\Delta t} \circ \cdots \circ V_{\Delta t}}_{n_{\text{sub}}\ \text{substeps}}(\mathbf{s}_t;\ \mathbf{a}_t) \tag{13}$$

The neural surrogate f_θ replaces this composition with a single forward pass: $f_\theta(\mathbf{z}_t) \approx T(\mathbf{s}_t, \mathbf{a}_t) - \mathbf{s}_t$.

6 Computation Within a Single Verlet Substep

Each Position Verlet substep $V_{\Delta t}$ is not a simple algebraic update — the velocity step (12b) requires evaluating the acceleration $\ddot{\mathbf{q}}(\mathbf{q}^{n+1/2})$, which triggers the full force assembly pipeline. The computation proceeds in three stages.

6.1 Stage 1: Half-Step Positions (Kinematic)

Advance generalized positions by half a timestep using current velocities:

$$x_i^{n+\frac{1}{2}} = x_i^n + \tfrac{1}{2}\Delta t\, \dot{x}_i^n, \qquad y_i^{n+\frac{1}{2}} = y_i^n + \tfrac{1}{2}\Delta t\, \dot{y}_i^n, \qquad i = 1, \ldots, 21 \tag{14a}$$

$$\psi_e^{n+\frac{1}{2}} = \psi_e^n + \tfrac{1}{2}\Delta t\, \omega_{z,e}^n, \qquad e = 1, \ldots, 20 \tag{14b}$$

This produces 62 updated position values (42 nodal + 20 element).

6.2 Stage 2: Force Assembly and Velocity Update (Dynamic)

Using the half-stepped configuration $\mathbf{q}^{n+1/2}$, the simulator computes all forces and accelerations. This is the expensive stage.

Step 2a: Geometry update. Recompute geometric quantities from the half-stepped positions:

$$\mathbf{t}_e = \frac{\mathbf{x}_{e+1}^{n+\frac{1}{2}} - \mathbf{x}_e^{n+\frac{1}{2}}}{\|\mathbf{x}_{e+1}^{n+\frac{1}{2}} - \mathbf{x}_e^{n+\frac{1}{2}}\|}, \qquad e = 1, \ldots, 20 \qquad \text{(tangent vectors)} \tag{15a}$$

$$\ell_e = \|\mathbf{x}_{e+1}^{n+\frac{1}{2}} - \mathbf{x}_e^{n+\frac{1}{2}}\|, \qquad e = 1, \ldots, 20 \qquad \text{(element lengths)} \tag{15b}$$

$$\varepsilon_e = \frac{\ell_e}{\ell_e^{\text{rest}}} \qquad \text{(dilatation / axial stretch)} \tag{15c}$$

Step 2b: Internal shear–stretch forces. Compute the generalized strain $\boldsymbol{\sigma}$ in the local (material) frame and apply the constitutive law:

$$\boldsymbol{\sigma}_e = \mathbf{Q}_e^T \, \frac{\mathbf{x}_{e+1}^{n+\frac{1}{2}} - \mathbf{x}_e^{n+\frac{1}{2}}}{\ell_e^{\text{rest}}} \qquad \text{(strain in local frame)} \tag{16a}$$

$$\mathbf{n}_e^L = \mathbf{S}\,(\boldsymbol{\sigma}_e - \boldsymbol{\sigma}_e^{\text{rest}}) \qquad \text{(internal stress, constitutive law)} \tag{16b}$$

$$\mathbf{F}_i^{\text{int}} = \Delta_s \left(\frac{\mathbf{Q}_e^T \, \mathbf{n}_e^L}{\varepsilon_e} \right) \qquad \text{(internal force per node, finite differences)} \tag{16c}$$

where $\mathbf{Q}_e$ is the director (rotation) matrix of element e and Δ_s denotes the discrete spatial derivative (difference kernel) on the staggered grid.

Step 2c: Internal bending–twist torques. Compute the curvature and bending–twist stresses:

$$\boldsymbol{\kappa}_v = \text{curvature at Voronoi nodes} \qquad \text{(from director differences)} \tag{17a}$$

$$\boldsymbol{\tau}_v^L = \mathbf{B}\,(\boldsymbol{\kappa}_v - \boldsymbol{\kappa}_v^{\text{rest}}) \qquad \text{(bending–twist couple, constitutive law)} \tag{17b}$$

The internal torque on each element e is then assembled from four contributions:

$$\mathbf{m}_e^{\text{int}} = \underbrace{\Delta_s\left(\frac{\boldsymbol{\tau}_v^L}{\hat{\varepsilon}_v^3}\right)}_{\text{bend–twist gradient}} + \underbrace{\mathcal{A}\left[\frac{(\boldsymbol{\kappa}_v \times \boldsymbol{\tau}_v^L)\,\hat{\ell}_v}{\hat{\varepsilon}_v^3}\right]}_{\text{curvature} \times \text{couple}} + \underbrace{(\mathbf{Q}_e\mathbf{t}_e) \times \mathbf{n}_e^L \cdot \ell_e^{\text{rest}}}_{\text{shear–stretch couple}} + \underbrace{\frac{\mathbf{J}_e\,\boldsymbol{\omega}_e}{\varepsilon_e} \times \boldsymbol{\omega}_e}_{\text{Lagrangian transport}} + \underbrace{\frac{\mathbf{J}_e\,\boldsymbol{\omega}_e}{\varepsilon_e^2}\,\dot{\varepsilon}_e}_{\text{unsteady dilatation}} \tag{18}$$

where $\hat{\varepsilon}_v$ is the Voronoi dilatation, $\mathcal{A}[\cdot]$ is the quadrature kernel, $\mathbf{J}_e$ is the second moment of inertia, and $\dot{\varepsilon}_e$ is the dilatation rate computed from position and velocity data.

Step 2d: External forces. Add the external distributed loads at each node or element:

$$\mathbf{f}_i^{\text{grav}} = m_i\,\mathbf{g} \qquad \text{(gravity, per node)} \tag{19a}$$

$$\mathbf{f}_i^{\text{rft}} = -c_t\,\mathbf{v}_{t,i} - c_n\,\mathbf{v}_{n,i} \qquad \text{(anisotropic ground friction, per node)} \tag{19b}$$

$$\mathbf{f}_i^{\text{damp}} = -\gamma\,\mathbf{v}_i \qquad \text{(numerical damping, per node)} \tag{19c}$$

Step 2e: Acceleration computation. Divide the total force by mass to obtain accelerations:

$$\ddot{x}_i = \frac{F_{x,i}^{\text{int}} + f_{x,i}^{\text{grav}} + f_{x,i}^{\text{rft}} + f_{x,i}^{\text{damp}}}{m_i}, \quad \ddot{y}_i = \frac{F_{y,i}^{\text{int}} + f_{y,i}^{\text{grav}} + f_{y,i}^{\text{rft}} + f_{y,i}^{\text{damp}}}{m_i} \tag{20a}$$

$$\dot{\omega}_{z,e} = \frac{1}{\varepsilon_e}\,\mathbf{J}_e^{-1}\left(\mathbf{m}_e^{\text{int}} + \mathbf{m}_e^{\text{ext}}\right)\Big|_z \tag{20b}$$

This yields 42 translational accelerations (21 nodes $\times$ 2 components) and 20 angular accelerations (20 elements), totaling 62 acceleration evaluations. These accelerations are *not* stored as state — they are recomputed from scratch at every substep because they depend on the half-stepped configuration $\mathbf{q}^{n+1/2}$.

Step 2f: Velocity update. Apply accelerations to advance velocities by a full timestep:

$$\dot{x}_i^{n+1} = \dot{x}_i^n + \Delta t\, \ddot{x}_i, \qquad \dot{y}_i^{n+1} = \dot{y}_i^n + \Delta t\, \ddot{y}_i \tag{21a}$$

$$\omega_{z,e}^{n+1} = \omega_{z,e}^n + \Delta t\, \dot{\omega}_{z,e} \tag{21b}$$

6.3 Stage 3: Complete Position Update (Kinematic)

Advance positions to the full timestep using the *updated* velocities:

$$x_i^{n+1} = x_i^{n+\frac{1}{2}} + \tfrac{1}{2}\Delta t\, \dot{x}_i^{n+1}, \quad y_i^{n+1} = y_i^{n+\frac{1}{2}} + \tfrac{1}{2}\Delta t\, \dot{y}_i^{n+1}, \qquad i = 1, \ldots, 21 \tag{22a}$$

$$\psi_e^{n+1} = \psi_e^{n+\frac{1}{2}} + \tfrac{1}{2}\Delta t\, \omega_{z,e}^{n+1}, \qquad e = 1, \ldots, 20 \tag{22b}$$

6.4 Equation Count per RL Step

Operation	**Per substep**	$\times 500$	**Reference**
Half-step positions (Stage 1)	62	31,000	(14)
Geometry update (tangents, lengths, dilatation)	60	30,000	(15)
Shear–stretch stress + internal forces	$\sim$120	60,000	(16)
Bending–twist stress + internal torques	$\sim$200	100,000	(17–18)
External forces (gravity, RFT, damping)	126	63,000	(19)
Accelerations (translational + angular)	62	31,000	(20)
Velocity update (Stage 2f)	62	31,000	(21)
Full-step positions (Stage 3)	62	31,000	(22)
Total scalar operations	$\sim\mathbf{754}$	$\sim\mathbf{377{,}000}$	

Table 1: Approximate equation count for one RL control step ($\Delta t_{\text{ctrl}} = 0.5\,\text{s}$). The neural surrogate replaces all $\sim$377,000 scalar operations with a single forward pass $f_\theta\colon \mathbb{R}^{129} \to \mathbb{R}^{124}$.

The dominant cost is the internal torque assembly (18), which involves five terms per element, each requiring cross products, matrix–vector products, and finite differences.

7 Matrix-Operation Form

Every operation in the Verlet substep is either a *matrix–vector product* (with a constant or state-dependent matrix), an *element-wise nonlinear function*, or a *Hadamard (element-wise)*

product. This section reformulates the full substep as a dataflow pipeline of such operations, making the input/output interface of each block explicit.

We define the following matrix operators. **Shape convention:** PyElastica stores all quantities as 3D vectors ($(3, N)$ or $(3, N_e)$ arrays) even for planar motion. The matrix sizes below reflect this 3D storage. However, for the 2D planar case ($z \equiv 0$), the effective degrees of freedom reduce to 2 translational components per node and 1 rotational component (ω_z) per element, giving the 124 scalar state variables of Section 3.

7.1 Constant Matrices (Independent of State)

Symbol	Name	Structure	Size
$\mathbf{D}$	Difference operator (node → element)	Bidiagonal: $D_{e,i} = \delta_{i,e+1} - \delta_{i,e}$	20×21
$\mathbf{D}^T$	Transpose difference (element → node)	Force assembly	21×20
$\mathbf{A}_v$	Voronoi averaging (element → Voronoi)	Tridiagonal: $\frac{1}{2}((\cdot)_e + (\cdot)_{e+1})$	19×20
$\mathbf{D}_v$	Voronoi difference (Voronoi → element)	Bidiagonal	20×19
$\mathbf{S}_{\text{blk}}$	Shear–stretch stiffness	Block-diagonal: $\text{diag}(\mathbf{S}_1, \ldots, \mathbf{S}_{20})$	60×60
$\mathbf{B}_{\text{blk}}$	Bend–twist stiffness	Block-diagonal: $\text{diag}(\mathbf{B}_1, \ldots, \mathbf{B}_{19})$	57×57
$\mathbf{M}$	Mass	Diagonal: $\text{diag}(m_1, \ldots, m_{21})$	21×21
$\mathbf{M}^{-1}$	Inverse mass	Diagonal: $\text{diag}(1/m_1, \ldots, 1/m_{21})$	21×21
$\mathbf{J}_{\text{blk}}^{-1}$	Inverse second moment of inertia	Block-diagonal	60×60

Table 2: Constant sparse matrices. Assembled once at initialization. Sizes reflect PyElastica's 3D storage; effective 2D sizes are $\mathbf{S}_{\text{blk}} \in \mathbb{R}^{40\times 40}$ (2 components $\times$ 20 elements), $\mathbf{B}_{\text{blk}} \in \mathbb{R}^{19\times 19}$ (scalar κ_z at 19 Voronoi nodes), $\mathbf{J}_{\text{blk}}^{-1} \in \mathbb{R}^{20\times 20}$ (scalar J_z per element).

The block entries of $\mathbf{S}_{\text{blk}}$, $\mathbf{B}_{\text{blk}}$, $\mathbf{M}$, and $\mathbf{J}_{\text{blk}}^{-1}$ are computed from the material parameters (Section 2.5) and the circular cross-section ($R = 0.001\,\text{m}$):

$$A = \pi R^2 \approx 3.14 \times 10^{-6}\,\text{m}^2, \quad I = \tfrac{\pi R^4}{4} \approx 7.85 \times 10^{-13}\,\text{m}^4, \quad J = \tfrac{\pi R^4}{2} \approx 1.57 \times 10^{-12}\,\text{m}^4 \tag{23a}$$

Shear–stretch stiffness block. Each $\mathbf{S}_e$ is identical (uniform rod). In 3D: $\mathbf{S}_e = \text{diag}(GA,\ GA,\ EA)$. In the effective 2D case (one shear + one axial component):

$$\mathbf{S}_e^{(2\text{D})} = \text{diag}(GA,\ EA) = \text{diag}(2.09,\ 6.28)\ \text{N} \tag{24}$$

Bending–twist stiffness block. In 2D, only the z-bending curvature matters, so each block reduces to a scalar:

$$B_v^{(2D)} = EI = 2 \times 10^6 \times 7.85 \times 10^{-13} \approx 1.57 \times 10^{-6}\ \mathrm{N\,m^2} \tag{25}$$

Lumped mass. Interior nodes carry the mass of two half-elements; boundary nodes carry one half-element:

$$m_i = \rho A \times \begin{cases} \frac{1}{2}\,\ell^{\text{rest}} = 9.42 \times 10^{-5}\ \mathrm{kg} & i = 1 \text{ or } 21 \\ \ell^{\text{rest}} = 1.88 \times 10^{-4}\ \mathrm{kg} & i = 2, \ldots, 20 \end{cases} \tag{26}$$

Inverse second moment of inertia. In 2D, the rotational inertia per element about the z-axis is:

$$J_{z,e} = \rho I\, \ell_e^{\text{rest}} = 1200 \times 7.85 \times 10^{-13} \times 0.05 \approx 4.71 \times 10^{-11}\ \mathrm{kg\,m^2} \tag{27}$$

so $\mathbf{J}_{\text{blk}}^{-1(2D)} = \mathrm{diag}(1/J_{z,1}, \ldots, 1/J_{z,20})$ with $1/J_{z,e} \approx 2.12 \times 10^{10}\ (\mathrm{kg\,m^2})^{-1}$.

7.2 State-Dependent Matrices

Symbol	Name	Depends on	Size
$\mathbf{Q}_{\text{blk}}(\boldsymbol{\psi})$	Rotation (material $\to$ lab frame)	ψ_e via $\cos\psi_e, \sin\psi_e$	40×40
$\mathrm{diag}(\boldsymbol{\varepsilon}^{-1})$	Inverse dilatation	$\mathbf{x}$ via element lengths	20×20
$\mathrm{diag}(\hat{\boldsymbol{\varepsilon}}_v^{-3})$	Inv. cube Voronoi dilatation	$\mathbf{x}$ via Voronoi lengths	19×19
$\mathbf{C}_{\text{rft}}(\mathbf{t})$	Friction projection	tangent $\mathbf{t}_e$ (anisotropy direction)	42×42

Table 3: State-dependent matrices. Entries are nonlinear functions of the current configuration, reassembled at every substep.

7.3 Reference (Rest) Configuration Constants

The following quantities describe the undeformed rod and are computed once at initialization from the material and geometric parameters in Section 2.5:

Symbol	Name	Value / definition	Shape
$\boldsymbol{\ell}^{\text{rest}}$	Rest element lengths	$\ell_e^{\text{rest}} = L/N_e = 0.05\,\text{m}$ (uniform)	$\mathbb{R}^{20}$
$\hat{\boldsymbol{\ell}}_v^{\text{rest}}$	Rest Voronoi lengths	$\hat{\ell}_v^{\text{rest}} = \frac{1}{2}(\ell_e^{\text{rest}} + \ell_{e+1}^{\text{rest}}) = 0.05\,\text{m}$	$\mathbb{R}^{19}$
$\boldsymbol{\sigma}^{\text{rest}}$	Rest strain	$\boldsymbol{\sigma}_e^{\text{rest}} = (0,\ 0,\ 1)^T$ (no shear, unit stretch)	$\mathbb{R}^{20\times 3}$
$\boldsymbol{\kappa}_v^{\text{rest}}$	Rest curvature	Set by CPG: $\kappa_e^{\text{rest}}(t)$ from (44)	$\mathbb{R}^{19}$
$\mathbf{g}$	Gravity vector	$(0,\ 0,\ -9.81)\,\text{m}\,\text{s}^{-2}$ (only z; zero in 2D plane)	$\mathbb{R}^3$

Table 4: Reference configuration constants. $\boldsymbol{\sigma}^{\text{rest}}$ and $\boldsymbol{\ell}^{\text{rest}}$ are fixed; $\boldsymbol{\kappa}_v^{\text{rest}}$ is the **control input** that changes at each RL step.

7.4 Element-Wise Nonlinear Functions

The following element-wise (pointwise) operations appear between matrix multiplications:

Notation	Operation	Input	Output
$\lVert\cdot\rVert$	Euclidean norm per element	$\Delta\mathbf{x}_e \in \mathbb{R}^2$	$\ell_e \in \mathbb{R}$
$(\cdot)^{-1}$	Reciprocal	ℓ_e or ε_e	$1/\ell_e$ or $1/\varepsilon_e$
$(\cdot)^{-3}$	Cube reciprocal	$\hat{\varepsilon}_v$	$\hat{\varepsilon}_v^{-3}$
$\cos(\cdot),\ \sin(\cdot)$	Trigonometric	ψ_e	Entries of $\mathbf{Q}_e$
$\times$	Cross product (bilinear)	$\mathbf{u}, \mathbf{w} \in \mathbb{R}^2$	$(\mathbf{u}\times\mathbf{w})_z \in \mathbb{R}$ (2D $\to$ scalar)
$\odot$	Hadamard product	Two vectors of same size	Element-wise product

Table 5: Element-wise nonlinear operations. Each is applied independently to every element or node.

7.5 The Full Substep as a Dataflow Pipeline

Using the notation above, one complete Position Verlet substep $V_{\Delta t}$ is the following pipeline. Each line shows: **output** $\leftarrow$ **operation(inputs)**, with the matrix or nonlinearity that produces it.

Stage 1: Kinematic half-step.

$$\boxed{\begin{array}{ll} \mathbf{x}^{n+\frac{1}{2}} \leftarrow \mathbf{x}^n + \frac{1}{2}\Delta t\ \dot{\mathbf{x}}^n & \text{[linear, per component]} \\ \boldsymbol{\psi}^{n+\frac{1}{2}} \leftarrow \boldsymbol{\psi}^n + \frac{1}{2}\Delta t\ \boldsymbol{\omega}^n & \text{[linear, per element]} \end{array}} \tag{28}$$

Input: $\mathbf{x}^n \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\psi}^n \in \mathbb{R}^{20}$, $\dot{\mathbf{x}}^n \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\omega}^n \in \mathbb{R}^{20}$ (124 scalars total).
Output: $\mathbf{x}^{n+1/2} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\psi}^{n+1/2} \in \mathbb{R}^{20}$ (62 scalars).

Stage 2a: Geometry.

$$\boxed{\begin{aligned} \Delta\mathbf{x} &\leftarrow \mathbf{D}\,\mathbf{x}^{n+\frac{1}{2}} && [\text{constant } \mathbf{D} \in \mathbb{R}^{20\times 21},\ \text{per component}] \\ \boldsymbol{\ell} &\leftarrow \|\Delta\mathbf{x}\|_{\text{row}} && [\text{element-wise norm}] \\ \boldsymbol{\varepsilon} &\leftarrow \boldsymbol{\ell} \oslash \boldsymbol{\ell}^{\text{rest}} && [\text{element-wise division}] \\ \mathbf{t} &\leftarrow \Delta\mathbf{x} \oslash \boldsymbol{\ell} && [\text{element-wise division (normalize)}] \end{aligned}} \tag{29}$$

Input: $\mathbf{x}^{n+1/2} \in \mathbb{R}^{21\times 2}$.
Output: $\Delta\mathbf{x} \in \mathbb{R}^{20\times 2}$, $\boldsymbol{\ell} \in \mathbb{R}^{20}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{20}$, $\mathbf{t} \in \mathbb{R}^{20\times 2}$.

Stage 2b: Rotation assembly.

$$\boxed{\mathbf{Q}_{\text{blk}} \leftarrow \bigoplus_{e=1}^{20} \begin{pmatrix} \cos\psi_e & -\sin\psi_e \\ \sin\psi_e & \cos\psi_e \end{pmatrix}} \tag{30}$$

Input: $\boldsymbol{\psi}^{n+1/2} \in \mathbb{R}^{20}$. *Output:* $\mathbf{Q}_{\text{blk}} \in \mathbb{R}^{40\times 40}$ (block-diagonal, $\cos$ / $\sin$ nonlinearity).

Stage 2c: Internal shear–stretch forces.

$$\boxed{\begin{aligned} \boldsymbol{\sigma} &\leftarrow \mathbf{Q}_{\text{blk}}^T\,(\Delta\mathbf{x} \oslash \boldsymbol{\ell}^{\text{rest}}) && [\text{state-dependent rotation}] \\ \mathbf{n}^L &\leftarrow \mathbf{S}_{\text{blk}}\,(\boldsymbol{\sigma} - \boldsymbol{\sigma}^{\text{rest}}) && [\text{constant stiffness}] \\ \mathbf{F}^{\text{int}} &\leftarrow \mathbf{D}^T\big[\operatorname{diag}(\boldsymbol{\varepsilon}^{-1})\ \mathbf{Q}_{\text{blk}}^T\,\mathbf{n}^L\big] && [\text{rotate, scale, difference}] \end{aligned}} \tag{31}$$

Input: $\Delta\mathbf{x} \in \mathbb{R}^{20\times 2}$, $\boldsymbol{\ell}^{\text{rest}} \in \mathbb{R}^{20}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{20}$, $\mathbf{Q}_{\text{blk}} \in \mathbb{R}^{40\times 40}$.
Output: $\boldsymbol{\sigma} \in \mathbb{R}^{20\times 2}$, $\mathbf{n}^L \in \mathbb{R}^{20\times 2}$, $\mathbf{F}^{\text{int}} \in \mathbb{R}^{21\times 2}$ (internal force per node).

Stage 2d: Internal bending–twist torques.

$$
\begin{aligned}
\boldsymbol{\kappa}_v &\leftarrow \text{curvature}(\mathbf{Q}_{\text{blk}}, \boldsymbol{\ell}_v^{\text{rest}}) && \text{[director differences at Voronoi]} \\
\boldsymbol{\tau}_v^L &\leftarrow \mathbf{B}_{\text{blk}}\,(\boldsymbol{\kappa}_v - \boldsymbol{\kappa}_v^{\text{rest}}) && \text{[constant stiffness]} \\
\mathbf{m}^{\text{int}} &\leftarrow \mathbf{D}_v\big[\text{diag}(\hat{\boldsymbol{\varepsilon}}_v^{-3})\,\boldsymbol{\tau}_v^L\big] && \text{[(i) bend–twist gradient]} \\
&\quad + \mathbf{A}_v\Big[(\boldsymbol{\kappa}_v \times \boldsymbol{\tau}_v^L) \odot \hat{\boldsymbol{\ell}}_v \odot \hat{\boldsymbol{\varepsilon}}_v^{-3}\Big] && \text{[(ii) curvature} \times \text{couple]} \\
&\quad + (\mathbf{Q}_{\text{blk}}\,\mathbf{t}) \times \mathbf{n}^L \odot \boldsymbol{\ell}^{\text{rest}} && \text{[(iii) shear–stretch couple]} \\
&\quad + (\mathbf{J}_{\text{blk}}\,\boldsymbol{\omega} \odot \boldsymbol{\varepsilon}^{-1}) \times \boldsymbol{\omega} && \text{[(iv) Lagrangian transport]} \\
&\quad + \mathbf{J}_{\text{blk}}\,\boldsymbol{\omega} \odot \boldsymbol{\varepsilon}^{-2} \odot \dot{\boldsymbol{\varepsilon}} && \text{[(v) unsteady dilatation]}
\end{aligned}
\tag{32}
$$

where the Voronoi dilatation and dilatation rate are intermediate quantities:

$$\hat{\boldsymbol{\varepsilon}}_v = \mathbf{A}_v\,\boldsymbol{\varepsilon} \in \mathbb{R}^{19} \qquad \text{(average of adjacent element dilatations)} \tag{33}$$

$$\dot{\boldsymbol{\varepsilon}} = \big(\Delta\dot{\mathbf{x}} \cdot \mathbf{t}\big) \oslash \boldsymbol{\ell}^{\text{rest}} \in \mathbb{R}^{20} \qquad \text{(from velocity and tangent projections)} \tag{34}$$

Input: $\mathbf{Q}_{\text{blk}} \in \mathbb{R}^{40\times 40}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{20}$, $\boldsymbol{\omega} \in \mathbb{R}^{20}$, $\mathbf{t} \in \mathbb{R}^{20\times 2}$, $\mathbf{n}^L \in \mathbb{R}^{20\times 2}$, $\dot{\mathbf{x}} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\ell}^{\text{rest}} \in \mathbb{R}^{20}$, $\hat{\boldsymbol{\ell}}_v^{\text{rest}} \in \mathbb{R}^{19}$.
Output: $\boldsymbol{\kappa}_v \in \mathbb{R}^{19}$, $\boldsymbol{\tau}_v^L \in \mathbb{R}^{19}$, $\hat{\boldsymbol{\varepsilon}}_v \in \mathbb{R}^{19}$, $\dot{\boldsymbol{\varepsilon}} \in \mathbb{R}^{20}$, $\mathbf{m}^{\text{int}} \in \mathbb{R}^{20}$.

Stage 2e: External forces and torques.

$$
\begin{aligned}
\mathbf{F}^{\text{ext}} &\leftarrow \underbrace{\mathbf{M}\,\mathbf{g}}_{\text{gravity}} - \underbrace{\mathbf{C}_{\text{rft}}(\mathbf{t})\,\dot{\mathbf{x}}}_{\text{anisotropic friction}} - \underbrace{\gamma\,\dot{\mathbf{x}}}_{\text{damping}} && \text{[translational]} \\
\mathbf{m}^{\text{ext}} &\leftarrow \mathbf{0} && \text{[no external torques in this model]}
\end{aligned}
\tag{35}
$$

Input: $\dot{\mathbf{x}} \in \mathbb{R}^{21\times 2}$, $\mathbf{t} \in \mathbb{R}^{20\times 2}$, $\mathbf{M} \in \mathbb{R}^{21\times 21}$, $\mathbf{g} \in \mathbb{R}^2$, $c_t, c_n, \gamma \in \mathbb{R}$.
Output: $\mathbf{F}^{\text{ext}} \in \mathbb{R}^{21\times 2}$, $\mathbf{m}^{\text{ext}} \in \mathbb{R}^{20}$.

Note: gravity acts only in the vertical direction. For planar motion on a horizontal surface, $\mathbf{g}_{\text{2D}} = (0, 0)$ and the gravitational effect enters only through the normal force that enables friction. The CPG muscle actuation enters via $\boldsymbol{\kappa}_v^{\text{rest}}$ in Stage 2d, not as an external torque.

The friction matrix $\mathbf{C}_{\text{rft}}(\mathbf{t})$ decomposes the velocity into tangential and normal components via the tangent vectors:

$$\mathbf{C}_{\text{rft}} = c_t\,\mathbf{P}_t + c_n\,(\mathbf{I} - \mathbf{P}_t), \qquad \mathbf{P}_t = \text{blkdiag}\big(\mathbf{t}_i\,\mathbf{t}_i^T\big)_{i=1}^N \tag{36}$$

Stage 2f: Accelerations.

$$\boxed{\begin{array}{ll} \ddot{\mathbf{x}} \leftarrow \mathbf{M}^{-1}\left(\mathbf{F}^{\text{int}} + \mathbf{F}^{\text{ext}}\right) & \text{[diagonal inverse mass]} \\ \dot{\boldsymbol{\omega}} \leftarrow \operatorname{diag}(\boldsymbol{\varepsilon})\, \mathbf{J}_{\text{blk}}^{-1}\left(\mathbf{m}^{\text{int}} + \mathbf{m}^{\text{ext}}\right) & \text{[block-diagonal inverse inertia]} \end{array}} \tag{37}$$

Input: $\mathbf{F}^{\text{int}} \in \mathbb{R}^{21\times 2}$, $\mathbf{F}^{\text{ext}} \in \mathbb{R}^{21\times 2}$, $\mathbf{m}^{\text{int}} \in \mathbb{R}^{20}$, $\mathbf{m}^{\text{ext}} \in \mathbb{R}^{20}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{20}$.
Output: $\ddot{\mathbf{x}} \in \mathbb{R}^{21\times 2}$, $\dot{\boldsymbol{\omega}} \in \mathbb{R}^{20}$ (62 scalars).

Stage 2g: Velocity update.

$$\boxed{\begin{array}{ll} \dot{\mathbf{x}}^{n+1} \leftarrow \dot{\mathbf{x}}^{n} + \Delta t\, \ddot{\mathbf{x}} & \text{[linear, per component]} \\ \boldsymbol{\omega}^{n+1} \leftarrow \boldsymbol{\omega}^{n} + \Delta t\, \dot{\boldsymbol{\omega}} & \text{[linear, per element]} \end{array}} \tag{38}$$

Input: $\dot{\mathbf{x}}^{n} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\omega}^{n} \in \mathbb{R}^{20}$, $\ddot{\mathbf{x}} \in \mathbb{R}^{21\times 2}$, $\dot{\boldsymbol{\omega}} \in \mathbb{R}^{20}$, $\Delta t \in \mathbb{R}$.
Output: $\dot{\mathbf{x}}^{n+1} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\omega}^{n+1} \in \mathbb{R}^{20}$ (62 scalars).

Stage 3: Kinematic full-step.

$$\boxed{\begin{array}{l} \mathbf{x}^{n+1} \leftarrow \mathbf{x}^{n+\frac{1}{2}} + \frac{1}{2}\Delta t\, \dot{\mathbf{x}}^{n+1} \\ \boldsymbol{\psi}^{n+1} \leftarrow \boldsymbol{\psi}^{n+\frac{1}{2}} + \frac{1}{2}\Delta t\, \boldsymbol{\omega}^{n+1} \end{array}} \tag{39}$$

Input: $\mathbf{x}^{n+1/2} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\psi}^{n+1/2} \in \mathbb{R}^{20}$, $\dot{\mathbf{x}}^{n+1} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\omega}^{n+1} \in \mathbb{R}^{20}$.
Output: $\mathbf{x}^{n+1} \in \mathbb{R}^{21\times 2}$, $\boldsymbol{\psi}^{n+1} \in \mathbb{R}^{20}$ (62 scalars).

7.6 Dataflow Summary

The complete dependency graph for one substep is:

$$\mathbf{s}^{n} \xrightarrow{\text{Stage 1}} \mathbf{q}^{n+\frac{1}{2}} \xrightarrow[\text{Geometry}]{\mathbf{D},\, \|\cdot\|} (\boldsymbol{\ell}, \boldsymbol{\varepsilon}, \mathbf{t}) \xrightarrow[\text{Rotation}]{\cos,\sin} \mathbf{Q}_{\text{blk}} \xrightarrow[\text{Constitutive}]{\mathbf{S}_{\text{blk}},\, \mathbf{B}_{\text{blk}}} (\mathbf{n}^{L}, \boldsymbol{\tau}^{L}) \xrightarrow[\text{Assembly}]{\mathbf{D}^{T},\, \times,\, \odot} (\mathbf{F}^{\text{int}}, \mathbf{m}^{\text{int}}) \tag{40}$$

$$(\mathbf{F}^{\text{int}} + \mathbf{F}^{\text{ext}},\ \mathbf{m}^{\text{int}} + \mathbf{m}^{\text{ext}}) \xrightarrow{\mathbf{M}^{-1},\, \mathbf{J}_{\text{blk}}^{-1}} (\ddot{\mathbf{x}}, \dot{\boldsymbol{\omega}}) \xrightarrow{+\Delta t} (\dot{\mathbf{x}}^{n+1}, \boldsymbol{\omega}^{n+1}) \xrightarrow{\text{Stage 3}} \mathbf{s}^{n+1} \tag{41}$$

Every arrow is either a sparse matrix multiplication (constant or state-dependent) or an element-wise nonlinearity. The **only** nonlinearities are: $\|\cdot\|$ (norms), $(\cdot)^{-1}$ (reciprocals), $\cos / \sin$ (rotations), and $\times$ (cross products). All other operations are linear in their inputs.

This structure is why the substep is amenable to neural network approximation: it is a sequence of linear maps interleaved with pointwise nonlinearities — structurally identical to a deep network with sparse weight matrices and specialized activation functions.

7.7 From Single Substep to the Full RL Step

The dataflow above describes a single Verlet substep $V_{\Delta t}$. One RL control step composes $n_{\text{sub}} = 500$ such substeps, but with an important detail: the rest curvature $\boldsymbol{\kappa}_v^{\text{rest}}$ is *not constant* across substeps. It is a time-varying signal generated by the CPG from the RL action $\mathbf{a}$.

Step 0: Action → CPG parameters. The RL policy outputs $\mathbf{a} = (A, f, k, \phi_0, b) \in \mathbb{R}^5$ once per control step. These five scalars parameterize the CPG traveling wave for the entire step duration $\Delta t_{\text{ctrl}} = 0.5\,\text{s}$. The action is held fixed; no new policy evaluation occurs until all 500 substeps complete.

Substep loop. For $n = 0, 1, \ldots, n_{\text{sub}} - 1$ (superscripts on t^n, $\mathbf{s}^n$, and $\boldsymbol{\kappa}^{\text{rest},\,n}$ denote the substep *index*, not exponentiation):

$$\boxed{\begin{aligned} t^n &\leftarrow t_{\text{step}} + n \cdot \Delta t && \text{[current substep time]} \\ \kappa_v^{\text{rest},\,n} &\leftarrow A \sin\!\big(2\pi k \cdot s_v + 2\pi f \cdot t^n + \phi_0\big) + b, \quad v = 1, \ldots, 19 && \text{[CPG evaluation at Voronoi nodes]} \\ \mathbf{s}^{n+1} &\leftarrow V_{\Delta t}\big(\mathbf{s}^n;\ \boldsymbol{\kappa}^{\text{rest},\,n}\big) && \text{[one Verlet substep (Sections 7.5–7.6)]} \end{aligned}} \tag{42}$$

The rest curvature enters the substep at Stage 2d (Eq. 32), where it modulates the bending–twist couple $\boldsymbol{\tau}_v^L = \mathbf{B}_{\text{blk}}\,(\boldsymbol{\kappa}_v - \boldsymbol{\kappa}_v^{\text{rest},\,n})$. Because the CPG frequency f produces a traveling wave, $\boldsymbol{\kappa}^{\text{rest},\,n}$ changes at every substep even though the action $\mathbf{a}$ is held fixed. The 500 substeps are therefore *not identical*: each sees a different rest curvature configuration, and the snake body responds dynamically to the propagating wave.

Transition operator dataflow. The full RL step, combining action decoding, CPG evaluation, and substep composition, is:

$$\mathbf{a}_t \xrightarrow{\text{CPG decode}} (A, f, k, \phi_0, b) \xrightarrow[\text{per substep}]{\boldsymbol{\kappa}^{\text{rest}}(t^n)} \underbrace{V_{\Delta t}\big(\,\cdot\,;\ \boldsymbol{\kappa}^{\text{rest},\,0}\big) \circ \cdots \circ V_{\Delta t}\big(\,\cdot\,;\ \boldsymbol{\kappa}^{\text{rest},\,499}\big)}_{\text{500 substeps, each with distinct } \boldsymbol{\kappa}^{\text{rest}}} \longrightarrow \mathbf{s}_{t+1} \tag{43}$$

Note that the composition is ordered: substep 0 applies first (innermost), substep 499 last (outermost). This is not a simple power iteration $V_{\Delta t}^{500}$ because the rest curvature changes each time.

	Quantity	Shape
Input	State $\mathbf{s}_t$: positions, velocities, yaw angles, angular velocities	$\mathbb{R}^{124}$
	Action $\mathbf{a}_t$: CPG parameters (A, f, k, ϕ_0, b)	$\mathbb{R}^5$
	Step start time t_{step} (for CPG phase)	$\mathbb{R}$
Output	Next state $\mathbf{s}_{t+1} = T(\mathbf{s}_t, \mathbf{a}_t)$	$\mathbb{R}^{124}$
Internal	CPG rest curvatures (recomputed each substep)	$\mathbb{R}^{19} \times 500$
	All intermediate quantities from Sections 7.5–7.6	—

Table 6: Input/output interface of the transition operator T. The neural surrogate $f_\theta \colon \mathbb{R}^{129} \to \mathbb{R}^{124}$ replaces this entire pipeline — 500 CPG evaluations and 500 Verlet substeps — with a single forward pass.

Input and output of $T(\mathbf{s}_t, \mathbf{a}_t)$.

Equation count (full RL step). Table 1 counts ∼377,000 scalar operations for the 500 Verlet substeps. Adding the CPG evaluation (19 sine evaluations $\times$ 500 substeps $=$ 9,500 additional transcendental operations) brings the total to approximately $\mathbf{386{,}500}$ scalar operations per RL step. The CPG cost is small relative to the force assembly, but the time-varying nature of $\boldsymbol{\kappa}^{\text{rest}}$ means the substeps are not interchangeable — each sees a different rest curvature. The surrogate must therefore learn the *integrated effect* of the entire traveling wave over Δt_{ctrl}, not just the response to a single static curvature configuration.

8 Control Input

The rest curvature $\boldsymbol{\kappa}_0(s, t)$ is not a system parameter — it is a **time-varying forcing** set by the RL policy at each control step via the CPG:

$$\kappa_e^{\text{rest}}(t) = A \sin\big(2\pi k \cdot s_e + 2\pi f \cdot t + \phi_0\big) + b \tag{44}$$

$$\mathbf{a} = (A,\ f,\ k,\ \phi_0,\ b) \in \mathbb{R}^5 \tag{45}$$

Symbol	Name	Range	Units
A	Amplitude	$[0, 5]$	$\mathrm{rad\,m^{-1}}$
f	Frequency	$[0.5, 3.0]$	Hz
k	Wave number	$[0.5, 3.5]$	wavelengths per body length
ϕ_0	Phase offset	$[0, 2\pi]$	rad
b	Turn bias	$[-2, 2]$	$\mathrm{rad\,m^{-1}}$

9 Complete System Summary

Component	**Specification**
$\mathcal{L}$	**Continuous:** Cosserat rod PDEs (2) with linear elastic constitutive law (3). **Semi-discrete:** 124 coupled first-order ODEs (10).
Ω	$s \in [0, 1.0\,\mathrm{m}] \times t \in [0, T]$. Discretized: $N = 21$ nodes, $N_e = 20$ elements, staggered grid.
BC	Free–free: $\mathbf{F} = \mathbf{M} = \mathbf{0}$ at both endpoints (5).
IC	Straight rod at rest: $\mathbf{x}(s, 0) = \mathbf{x}_0 + s\hat{\mathbf{e}}_x$, $\mathbf{v} = \boldsymbol{\kappa} = \boldsymbol{\omega} = \mathbf{0}$ (6).
$\boldsymbol{\theta}$	$\{E = 2{\times}10^6\,\mathrm{Pa},\ \nu = 0.5,\ \rho = 1200\,\mathrm{kg/m^3},\ R = 0.001\,\mathrm{m},\ L = 1.0\,\mathrm{m}\}$ $\{c_t = 0.01,\ c_n = 0.1,\ \gamma = 0.1\}$ $\{\Delta t = 0.001\,\mathrm{s},\ n_{\mathrm{sub}} = 500,\ \text{Position Verlet}\}$
Control	$\mathbf{a} = (A, f, k, \phi_0, b) \in \mathbb{R}^5$ via CPG (44); enters as $\boldsymbol{\kappa}_0(s, t)$.